



ALEXIS GRÉDELUE

Master's Student in Universe & Particle Physics

Scientific curiosity • Analytical thinking • Research autonomy • Collaborative research • Technical communication

Motivated by a strong curiosity for both fundamental physics and artificial intelligence, I am eager to pursue a PhD at the intersection of these fields. My research experiences in cosmology, high-energy physics, and machine learning have strengthened my desire to contribute to the development of advanced data-driven methods for scientific discovery. I am particularly interested in applying AI techniques to complex physics problems, from signal analysis to large-scale data interpretation. Through a PhD, I aim to deepen my theoretical and technical expertise while contributing to innovative research in modern physics and machine learning.

PERSONAL INFORMATION

 +33 6 76 99 16 23

 alexis.gre74@gmail.com

 Location: Annecy, France

 [Linkedin](#)

 [Github](#)

LANGUAGES

French: Native

English: Professional Working Proficiency (CEFR B2)

Japanese: Basics

PROFESSIONAL EXPERIENCE

Research intern in ATLAS team of CERN

LPCA Clermont-Ferrand, France | Feb 2026 - July 2026

Research internship at Laboratoire de Physique de Clermont Auvergne focused on the application of machine learning techniques to the study of Axion-Like Particles (ALPs) within the ATLAS Collaboration at CERN. Developed and evaluated machine learning approaches for the analysis and reinterpretation of ATLAS physics searches. Worked in a multidisciplinary research environment combining high-energy physics, data analysis, and artificial intelligence.

Key Responsibilities

- Optimized machine learning models, including Boosted Decision Trees (BDTs) and neural networks, for the classification of particle-physics signals associated with ALP searches.
- Analyzed Monte Carlo simulation datasets and interpreted event kinematics used in ATLAS analyses.
- Presented research progress and technical updates to the LPCA ATLAS research team every two weeks.

Key Achievements

- Developed ML pipelines using PyTorch, XGBoost and scikit-learn to classify ALP signal regions from Monte Carlo event kinematics and estimate signal yields for reinterpretation studies.
- Contributed to the machine-learning-based reinterpretation of ATLAS searches for Axion-Like Particles in collaboration with PhD researchers from the ATLAS team.

Research intern in Cosmology

Kumamoto University, Japan | April 2024 - July 2024

Conducted a bibliographic review on the foundations of cosmology (redshift, Hubble's law, Friedmann equations) and performed data analysis on Type Ia supernovae. Used astronomical datasets to estimate theoretical energy density components by minimizing residuals with observational data, providing independent evidence supporting the existence of dark energy.

Key responsibilities

- Programmed and analyzed data using Python
- Performed data analysis on experimental Type Ia supernova datasets
- Attended several conferences on astronomy
- Delivered weekly oral presentations summarizing progress
- Collaborated with another student from the university on the same research topic

Key achievements

- Acquired a solid understanding of the fundamentals of cosmology
- Analyzed Type Ia supernova data to validate the existence of dark energy through confidence ellipse regions.

EDUCATION

Master's Degree in Universe & Particle Physics [EUPI Clermont-Ferrand, France | Sep 2025 - Aug 2026](#)

Relevant coursework: Cosmology (17/20), General Relativity, Introduction to Particle Physics, Symmetries, Quantum Field Theory, Electroweak Physics, Neutrino Physics, Detectors, QCD, QED

University Diploma in Data Science [EUPI Clermont-Ferrand, France | Sep 2025 - Aug 2026](#)

Relevant coursework: Data analysis using Python (16.3/20), Machine Learning (14.75/20), Deep Learning (13.3/20), Quantum computing, Statistics

Master of Engineering Physics [Polytech Clermont-Ferrand, France | Sep 2022 - Aug 2025](#)

Relevant coursework: X-Ray Diffraction, Crystallography, Computer science, Fluid dynamics, Optics, Quantum physics, Statistical and Solid State Physics, Mechanical Design, Heat transfer phenomena

1st Prize — Clermont Auvergne PEPITE Hackathon (2024) [Clermont-Ferrand, France | May 2024](#)

Awarded first place in a multidisciplinary innovation challenge focused on sustainable resource management and water conservation solutions.

TECHNOLOGICAL SKILLS

Physics & Scientific Domains

High-Energy Physics, Cosmology, Experimental Particle Physics, Detector-based Data Analysis, Statistical Physics, Materials Science, Heat transfer phenomena, Experimental Physics.

Programming & Scientific Computing

Python (NumPy, SciPy, Pandas, Matplotlib, PyTorch, scikit-learn), Java.
Scientific Computing, Numerical Simulations, Monte Carlo Data Analysis, Statistical Analysis.

Machine Learning & AI

Machine Learning, Deep Learning, Neural Networks, Boosted Decision Trees (RandomForest, XGBoost), Classification & Signal Detection.
Feature Engineering, Data Preprocessing.

Software & Tools

Git, LaTeX, Visual Studio Code, n8n, Ollama.

Research & Technical Skills

Scientific Literature Review, Technical Reporting (LaTeX) & Oral Presentations.
Research Collaboration in International Teams, Experimental & Observational Data Interpretation.

PUBLICATIONS & RESEARCH CONTRIBUTIONS

Research intern in ATLAS team of CERN [LPCA Clermont-Ferrand, France | Feb 2026 - July 2026](#)

Ongoing contribution to ML reinterpretation studies within the ATLAS collaboration: internal research presentations on machine learning based reinterpretation for axion-like particle searches.

INDEPENDENT PROJECT

Java-based Game Mod Development (Personal Project) [February 2025 - Now](#)

Designed gameplay systems and custom mechanics for a large-scale sandbox game modification project involving modular systems implemented in Java.

Personal Research on Modular AI Architectures [March 2026 - Now](#)

Ongoing exploration of lightweight and energy-efficient AI architectures, including modular multi-agent systems for scientific reasoning and automation tasks.

REFERENCES

Prof. Keitaro Takahashi 高橋慶太郎, Professor, Kumamoto University (Japan)

Email: keitaro@kumamoto-u.ac.jp

Dr. Louie D. Corpe, Junior Professor, LPCA, ATLAS Collaboration (CERN)

Email: l.corpe@cern.ch